Network Fundamentals

From concepts to commands

1. What Is a Computer Network?

A group of two or more interconnected devices (computers, servers, printers, phones) that can share data, resources,hardware and services with each other.

Resource Sharing	Printers, files, internet connection shared across devices
Communication	Email, chat, video calls between connected devices
Centralized Data	Store and manage data in one place (servers)
Scalability	Easily add new devices as needs grow

2. Network Terminology

IP Address	Logical address (e.g. 192.168.1.10) identifying a device on a network — assigned, can change IP address is 32-bit divided into 4 octets each octet 8 bit is separated by dot . Class A, B, C
MAC Address	Physical address burned into the NIC (e.g. 00:1A:2B:3C:4D:5E) — unique, permanent Mac address is 48-bit written as 12 hexadecimal digits is separated by a colon :
NIC	Network Interface Card — hardware that connects a device to a network
Protocol	Set of rules devices follow to communicate (e.g. TCP, HTTP, DHCP, DNS)

3. Network Devices

Hub

Broadcasts data to all ports — outdated, no intelligence

Switch

Layer 2 — forwards frames using MAC address tables

Router

Layer 3 — forwards packets between different networks

Firewall

Filters traffic based on security rules

Access Point

Connects wireless clients to the wired network

Modem

Converts signals between the LAN and the ISP link

4. Network Models

OSI Model (7 layers)

7. Application
6. Presentation
5. Session
4. Transport
3. Network
2. Data Link
1. Physical

TCP/IP Model (4 layers)

Application

Transport

Internet

Network Access

4. Network Models

Peer-To-Peer

- No dedicated resources to present specific service.
- Easy to work with.
- All nodes are the same (equal to use the resources).
- Example : Windows Workgroup

Client - Server

- Some nodes (SERVER) are dedicated to present services to other nodes (CLIENTS).
- Server is more powerful.
- Examples: Mail Server, Web Server, File Server

5. Basic Network Commands

`ping`

Tests connectivity to another device
-n to set number of requests

`ipconfig / ifconfig`

Shows/manages local IP configuration

`tracert / traceroute`

Shows the path packets take to a destination

`nslookup`

Queries DNS to resolve domain names to IPs

`arp`

Shows/manages the IP-to-MAC address table

`netstat`

Shows active connections and listening ports

6. Assigning IP Addresses

Manual (Static)

Administrator manually types IP, subnet mask, gateway, DNS into the device's network settings. Doesn't change — used for servers, printers.

DHCP (Automatic)

A DHCP server automatically leases an IP address (and other settings) to the device when it joins the network. Most common method today.

APIPA

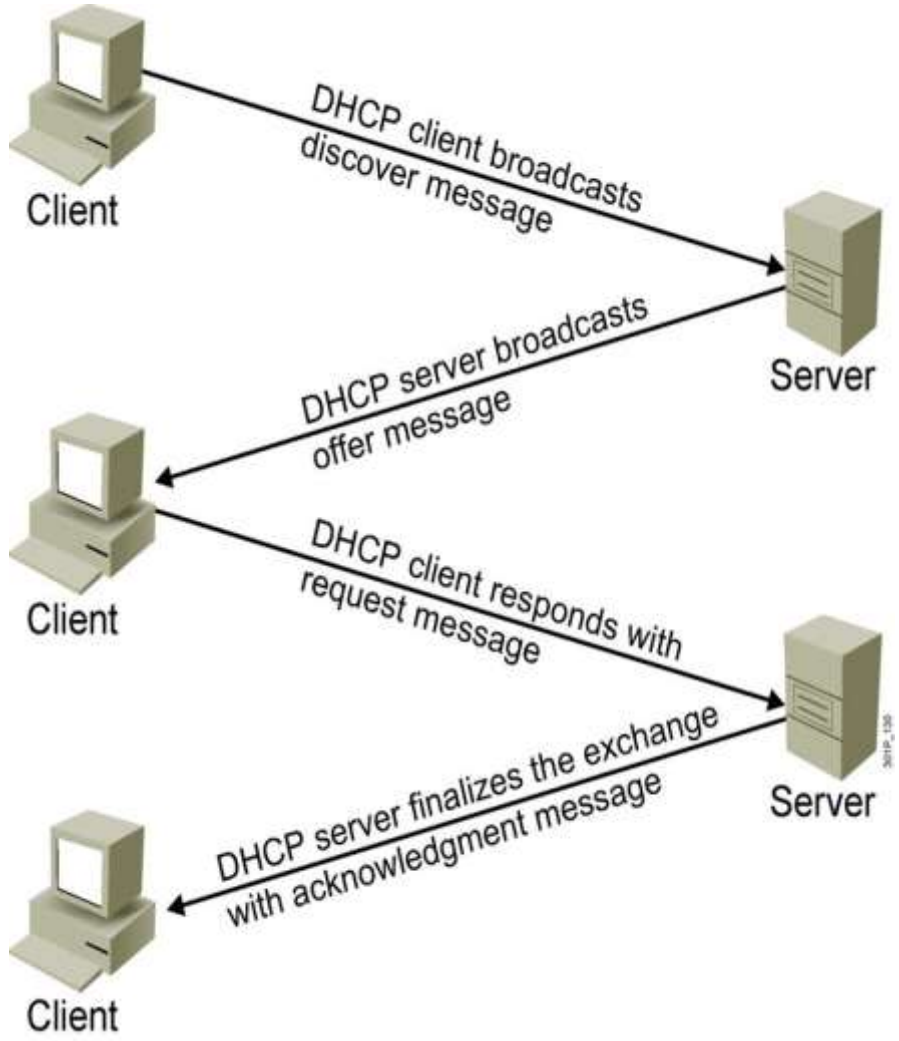
Automatic Private IP Addressing — if no DHCP server is found, Windows self-assigns an address in 169.254.0.0/16 so local devices can still talk to each other.

6. Assigning IP Addresses

DHCP (Automatic)

used to assign dynamically the IP Configuration including (Host IP, Subnet mask, Default Gateway, DNS server IP and the Lease Time).

Dora(discover –offer-request-acknowledgment)



7. The ipconfig Command

Windows command to view and manage IP configuration
(Linux/Mac equivalent: ifconfig or ip addr).

`ipconfig`	Shows basic IP, subnet mask, and default gateway
`ipconfig /all`	Shows full details: MAC address, DHCP server, DNS, lease times
`ipconfig /release`	Releases the current DHCP-assigned IP address
`ipconfig /renew`	Requests a new IP address from the DHCP server
`ipconfig /flushdns`	Clears the local DNS resolver cache



8. The nslookup Command

Queries DNS servers to resolve domain names to IP addresses (and vice versa) — useful for troubleshooting DNS issues.

`nslookup google.com`	Resolves the domain to its IP address(es)
`nslookup 8.8.8.8`	Reverse lookup — finds the domain name for an IP
`nslookup`	Enters interactive mode for multiple queries

9. The arp Command

ARP (Address Resolution Protocol) maps IP addresses to MAC addresses on a local network. The arp command lets you view and manage this table.

`arp -a`	Displays the current ARP table (IP-to-MAC mappings)
`arp -d *`	Deletes all entries from the ARP cache
`arp -s <ip> <mac>`	Manually adds a static ARP entry